



CSE6324-SOFTWARE VERIFICATION & VALIDATION

Topic: Hospital Management System

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Test Procedure Specification

Hospital Management System (HMS)

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Table of Contents

Document Control.....	3
Version History	4
Distribution List	4
1.0 Introduction	4
1.1 Purpose	5
1.2 Scope	5
1.3 References	7
2.0 Test Procedure	7
2.1 User Authentication Test Procedures	8

2.1.1 F001 Register Account Test Procedure	8
2.1.2 F002 Login Test Procedure	8
2.1.3 F003 Logout Test Procedure	9
2.2 Patient Functions Test Procedures	10
2.2.1 F004 Book Appointment Test Procedure	10
2.2.2 F005 View Appointment History Test Procedure	10
2.2.3 F006 Cancel Appointment Test Procedure	11
2.2.4 F007 View Prescription Test Procedure	11
2.2.5 F008 Submit Contact Query Test Procedure	12
2.3 Doctor Functions Test Procedures.....	12
2.3.1 F009 View Appointment Test Procedure.....	13
2.3.2 F010 Prescribe Medication Test Procedure	13
2.3.3 F011 Cancel Appointment Test Procedure	14
2.3.4 F012 View Prescription List Test Procedure	14
2.4 Admin Functions Test Procedure	15
2.4.1 F013 Add Doctor Test Procedure	16
2.4.2 F014 Delete Doctor Test Procedure	16
2.4.3 F015 View & Search Doctor List Test Procedure.....	17
2.4.4 F016 View & Search Patient List Test Procedure	17
2.4.5 F017 View & Search Appointment Details Test Procedure	18
2.4.6 F018 View Prescription List Test Procedure	18
2.4.7 F019 View & Search User Queries Test Procedure.....	19

1.0 Introduction

1.1 Purpose

This document outlines the procedures to be followed for the tests discussed in the Test Design Specification and elaborated upon in the Test Case Specification for the Hospital Management System (HMS). It outlines a methodical and consistent way for executing the identified HMS test cases.

For Test Design Specification and Test Cases, please refer to the following documents:

- i. HMS Test Design Specification, HMS_TDS
- ii. HMS Test Case Specification, HMS_TC

1.2 Scope

This Test Procedure Specification covers the execution procedures for the major functional features of the Hospital Management System (HMS). All procedures described in this document are based on the approved Software Requirements Specification (SRS), Test Design Specification (TDS) and Test Case Specification, and should be used to run the associated HMS Test Cases within the local test environment.

The features covered in this procedure include:

- i. User Authentication
 - F001 Register Account
 - F002 Login
 - F003 Logout
- ii. Patient Functions
 - F004 Book Appointment
 - F005 View Appointment History
 - F006 Cancel Appointment
 - F007 View Prescription
 - F008 Submit Contact Query
- iii. Doctor Functions
 - F009 View Appointment
 - F010 Prescribe Medication
 - F011 Cancel Appointment
 - F012 View Prescription List
- iv. Admin Functions

- F013 Add Doctor
- F014 Delete Doctor
- F015 View & Search Doctor List
- F016 View & Search Patient List
- F017 View & Search Appointment Details
- F018 View Prescription List
- F019 View & Search User Queries

This document is focusing on what is to be undertaken in preparation for each test case, which test cases to run and what to follow up after running each test case. Refreshed the test design and test case documents to indicate the detailed test design logic, without changing anything else in the document.

1.3 References

The following documents provide the test basis for this test design:

- i. HMS Software Requirement Specification (SRS), HMS_SRS
- ii. HMS Test Plan, HMS_TP
- iii. HMS Test Design Specifications, HMS_TDS

2.0 Test Procedure

2.1 User Authentication Test Procedures

2.1.1 F001 Register Account Test Procedure

Test Procedure ID	TPS-01-001
Objective	Execute the register account procedure for validating patient registration input and successful patient account creation.
Test Cases To Be Executed	TC-01-001, TC-01-002, TC-01-003, TC-01-004
Set Up	1. Launch Google Chrome browser. 2. Start the HMS local environment using XAMPP. 3. Open the HMS home page. 4. Ensure no patient user is currently logged into the system. 5. Prepare valid and invalid registration data for execution.
Wrap Up	1. Record the execution result and supporting evidence. 2. Reset or remove created test account data if necessary. 3. Close the browser.

2.1.2 F002 Login Test Procedure

Test Procedure ID	TPS-02-001
Objective	Execute the login procedure for verifying successful and unsuccessful login attempts for patient, doctor, and admin users.
Test Cases To Be Executed	TC-02-001, TC-02-002, TC-02-003, TC-02-004
Set Up	1. Launch Google Chrome browser. 2. Start the HMS local environment using XAMPP. 3. Ensure valid patient, doctor, and admin accounts are available. 4. Open the relevant HMS login page for the selected role.

	5. Prepare valid and invalid login credentials for execution.
Wrap Up	1. Logout from the system if login is successful. 2. Record the execution result and supporting evidence. 3. Close the browser.

2.1.3 F003 Logout Test Procedure

Test Procedure ID	TPS-03-001
Objective	Execute the logout procedure for verifying session termination and post-logout access control for authenticated HMS users.
Test Cases To Be Executed	TC-03-001, TC-03-002, TC-03-003, TC-03-004
Set Up	1. Launch Google Chrome browser. 2. Start the HMS local environment using XAMPP. 3. Log into HMS using a valid patient, doctor, or admin account. 4. Navigate to the corresponding dashboard page. 5. Prepare browser access for post-logout verification.
Wrap Up	1. Confirm that the session has ended successfully 2. Record the execution result and supporting evidence. 3. Close the browser.

2.2 Patient Functions Test Procedures

2.2.1 F004 Book Appointment Test Procedure

Test Procedure ID	TPS-04-001
Objective	Execute the book appointment procedure for verifying valid booking flow and major booking validation scenarios.
Test Cases To Be Executed	TC-04-001, TC-04-002, TC-04-003, TC-04-004
Set Up	1. Launch Google Chrome browser. 2. Start the HMS local environment using XAMPP. 3. Log into HMS using a valid patient account. 4. Navigate to the Book Appointment page. 5. Prepare valid and invalid appointment input data, including doctor availability and date/time conditions.
Wrap Up	1. Record the execution result and supporting evidence. 2. Remove or reset created appointment records if necessary. 3. Close the browser.

2.2.2 F005 View Appointment History Test Procedure

Test Procedure ID	TPS-05-001
Objective	Execute the appointment history procedure for verifying display of patient appointment records and status handling.
Test Cases To Be Executed	TC-05-001, TC-05-002, TC-05-003, TC-05-004
Set Up	1. Launch Google Chrome browser. 2. Start the HMS local environment using XAMPP.

	3. Log into HMS using a valid patient account. 4. Prepare patient accounts with existing appointments, no appointments, active appointments, and cancelled appointments. 5. Navigate to the Appointment History page.
Wrap Up	1. Record the execution result and supporting evidence. 2. Restore any modified appointment status if required. 3. Close the browser.

2.2.3 F006 Cancel Appointment Test Procedure

Test Procedure ID	TPS-06-001
Objective	Execute the patient appointment cancellation procedure for verifying successful cancellation and key restriction scenarios.
Test Cases To Be Executed	TC-06-001, TC-06-002, TC-06-003, TC-06-004
Set Up	1. Launch Google Chrome browser. 2. Start the HMS local environment using XAMPP. 3. Log into HMS using a valid patient account. 4. Prepare at least one active appointment and one cancelled appointment for the patient. 5. Navigate to the Appointment History page.
Wrap Up	1. Record the execution result and supporting evidence. 2. Reset cancelled appointment status if necessary for repeated testing. 3. Close the browser.

2.2.4 F007 View Prescription Test Procedure

Test Procedure ID	TPS-07-001
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Objective	Execute the patient prescription procedure for verifying prescription display, bill payment flow, and PDF bill generation.
Test Cases To Be Executed	TC-07-001, TC-07-002, TC-07-003, TC-07-004
Set Up	1. Launch Google Chrome browser. 2. Start the HMS local environment using XAMPP. 3. Log into HMS using a valid patient account. 4. Prepare patient records with existing prescriptions, no prescriptions, paid bill status, and unpaid bill status.
Wrap Up	1. Record the execution result and supporting evidence. 2. Save generated PDF evidence if applicable. 3. Reset payment-related test data if necessary. 4. Close the browser.

2.2.5 F008 Submit Contact Query Test Procedure

Test Procedure ID	TPS-08-001
Objective	Execute the contact query submission procedure for verifying successful message submission and validation handling
Test Cases To Be Executed	TC-08-001, TC-08-002, TC-08-003, TC-08-004
Set Up	1. Launch Google Chrome browser. 2. Start the HMS local environment using XAMPP. 3. Open the HMS Contact page. 4. Prepare valid and invalid contact query input data. 5. Ensure the query submission form is accessible.
Wrap Up	1. Record the execution result and supporting evidence. 2. Remove or reset submitted test query data if necessary. 3. Close the browser.

2.3 Doctor Functions Test Procedures

2.3.1 F009 View Appointment Test Procedure

Test Procedure ID	TPS-09-001
Objective	Execute the doctor appointment viewing procedure for verifying appointment list display and search behavior.
Test Cases To Be Executed	TC-09-001, TC-09-002, TC-09-003, TC-09-004
Set Up	1. Launch Google Chrome browser. 2. Start the HMS local environment using XAMPP. 3. Log into HMS using a valid doctor account. 4. Prepare doctor accounts with assigned appointments and no appointments. 5. Prepare valid and invalid patient contact numbers for search. 6. Navigate to the View Appointment page.
Wrap Up	1. Record the execution result and supporting evidence. 2. Clear search input if necessary. 3. Close the browser.

2.3.2 F010 Prescribe Medication Test Procedure

Test Procedure ID	TPS-10-001
Objective	Execute the prescribe medication procedure for verifying successful prescription creation and major validation scenarios.
Test Cases To Be Executed	TC-10-001, TC-10-002, TC-10-003, TC-10-004
Set Up	1. Launch Google Chrome browser. 2. Start the HMS local environment using XAMPP. 3. Log into HMS using a valid doctor account.

	4. Prepare at least one active appointment assigned to the logged-in doctor. 5. Prepare valid and invalid prescription input data. 6. Navigate to the Prescribe Medication interface from the appointment list.
Wrap Up	1. Record the execution result and supporting evidence. 2. Remove or reset created prescription data if necessary. 3. Close the browser.

2.3.3 F011 Cancel Appointment Test Procedure

Test Procedure ID	TPS-11-001
Objective	Execute the doctor appointment cancellation procedure for verifying successful cancellation and access restriction scenarios.
Test Cases To Be Executed	TC-11-001, TC-11-002, TC-11-003, TC-11-004
Set Up	1. Launch Google Chrome browser. 2. Start the HMS local environment using XAMPP. 3. Log into HMS using a valid doctor account. 4. Prepare at least one active appointment assigned to the doctor and one appointment not assigned to the doctor. 5. Navigate to the View Appointment page.
Wrap Up	1. Record the execution result and supporting evidence. 2. Reset appointment status if necessary for repeated testing. 3. Close the browser.

2.3.4 F012 View Prescription List Test Procedure

Test Procedure ID	TPS-12-001
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Objective	Execute the doctor prescription list procedure for verifying prescription record display and access restriction behavior.
Test Cases To Be Executed	TC-12-001, TC-12-002, TC-12-003, TC-12-004
Set Up	1. Launch Google Chrome browser. 2. Start the HMS local environment using XAMPP. 3. Log into HMS using a valid doctor account. 4. Prepare doctor accounts with existing prescription records and no prescription records. 5. Prepare at least one prescription record belonging to another doctor for restriction checking. 6. Navigate to the Prescription List page.
Wrap Up	1. Record the execution result and supporting evidence. 2. Close the browser.

2.4 Admin Functions Test Procedure

2.4.1 F013 Add Doctor Test Procedure

Test Procedure ID	TPS-13-001
Objective	Execute the add doctor procedure for verifying successful doctor account creation and key input validation scenarios.
Test Cases To Be Executed	TC-13-001, TC-13-002, TC-13-003, TC-13-004
Set Up	1. Launch Google Chrome browser. 2. Start the HMS local environment using XAMPP. 3. Log into HMS using a valid admin account. 4. Navigate to the Add Doctor page. 5. Prepare valid and invalid doctor account input data.
Wrap Up	1. Record the execution result and supporting evidence. 2. Remove or reset created doctor records if necessary. 3. Close the browser.

2.4.2 F014 Delete Doctor Test Procedure

Test Procedure ID	TPS-14-001
Objective	Execute the delete doctor procedure for verifying successful deletion and invalid deletion handling.
Test Cases To Be Executed	TC-14-001, TC-14-002, TC-14-003
Set Up	1. Launch Google Chrome browser. 2. Start the HMS local environment using XAMPP. 3. Log into HMS using a valid admin account. 4. Prepare at least one existing doctor record for deletion. 5. Navigate to the Doctor List or delete doctor function page.
Wrap Up	1. Record the execution result and supporting evidence. 2. Restore deleted doctor data if needed for repeated testing.

	3. Close the browser.
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2.4.3 F015 View & Search Doctor List Test Procedure

Test Procedure ID	TPS-15-001
Objective	Execute the doctor list procedure for verifying doctor record display and search behavior.
Test Cases To Be Executed	TC-15-001, TC-15-002, TC-15-003, TC-15-004
Set Up	1. Launch Google Chrome browser. 2. Start the HMS local environment using XAMPP. 3. Log into HMS using a valid admin account. 4. Prepare existing doctor records with known email values. 5. Navigate to the Doctor List page.
Wrap Up	1. Record the execution result and supporting evidence. 2. Clear search values if necessary. 3. Close the browser.

2.4.4 F016 View & Search Patient List Test Procedure

Test Procedure ID	TPS-16-001
Objective	Execute the patient list procedure for verifying patient record display and search behavior.
Test Cases To Be Executed	TC-16-001, TC-16-002, TC-16-003, TC-16-004
Set Up	1. Launch Google Chrome browser. 2. Start the HMS local environment using XAMPP. 3. Log into HMS using a valid admin account. 4. Prepare existing patient records with known contact numbers 5. Navigate to the Patient List page.
Wrap Up	1. Record the execution result and supporting evidence.

	2. Clear search values if necessary. 3. Close the browser.
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2.4.5 F017 View & Search Appointment Details Test Procedure

Test Procedure ID	TPS-17-001
Objective	Execute the appointment details procedure for verifying appointment record display and search behavior.
Test Cases To Be Executed	TC-17-001, TC-17-002, TC-17-003, TC-17-004
Set Up	1. Launch Google Chrome browser. 2. Start the HMS local environment using XAMPP. 3. Log into HMS using a valid admin account. 4. Prepare appointment records with known patient contact numbers. 5. Navigate to the Appointment Details page.
Wrap Up	1. Record the execution result and supporting evidence. 2. Clear search values if necessary. 3. Close the browser.

2.4.6 F018 View Prescription List Test Procedure

Test Procedure ID	TPS-18-001
Objective	Execute the admin prescription list procedure for verifying prescription record display and empty-list handling.
Test Cases To Be Executed	TC-18-001, TC-18-002, TC-18-003
Set Up	1. Launch Google Chrome browser. 2. Start the HMS local environment using XAMPP. 3. Log into HMS using a valid admin account. 4. Prepare prescription records or an empty prescription dataset as required.

	5. Navigate to the Prescription List page.
Wrap Up	1. Record the execution result and supporting evidence. 2. Close the browser.

2.4.7 F019 View & Search User Queries Test Procedure

Test Procedure ID	TPS-19-001
Objective	Execute the user query management procedure for verifying query display and search behavior.
Test Cases To Be Executed	TC-19-001, TC-19-002, TC-19-003, TC-19-004
Set Up	1. Launch Google Chrome browser. 2. Start the HMS local environment using XAMPP. 3. Log into HMS using a valid admin account. 4. Prepare submitted query records with known contact numbers. 5. Navigate to the User Queries page.
Wrap Up	1. Record the execution result and supporting evidence. 2. Clear search values if necessary. 3. Close the browser.